

# CSCI 48700: Artificial Intelligence

## Problem Set 2

Instructor: Dr. Snehasis Mukhopadhyay

Due date: 09/30/2025

1. Exercise 9.4 from the text-book, third edition (9.4 in second edition) –photocopy attached.

30 points

2. Exercise 9.6 from the text-book, third edition (9.9 in second edition)-photocopy attached.

30 points

3. Consider the following sentences:

John likes all kinds of food.

Apples are food.

Chicken is food.

Anything anyone eats and isn't killed by is food.

Bill eats peanuts and is still alive.

Sue eats everything Bill eats.

(a) Translate these sentences into formulas in predicate logic.

(b) Convert the formulas in (a) into clause form.

(c) Prove that John likes peanuts using resolution.

40 points